

Imagine...

V-CL Echo Pulse

Voicing the Voiceless



Atharva Vaze

Lead of Machine
Learning



Jayden Chiou

Lead of Outreach &
Growth



Craig de Fiesta

Lead of Hardware
Innovation



James Zhou

Lead of Financial
Operations



Idhant Ranjan

Lead of Data Science

5M Americans have lost their voice

They can think clearly, but can't speak.

The brain still works, but the connection is severed.

Current solutions are inadequate.

The consequences are immense.



Ye-Ye

James' Grandfather, ALS
Survivor, Inspiration for
VOCL

The VOCL Echo

Our Solution



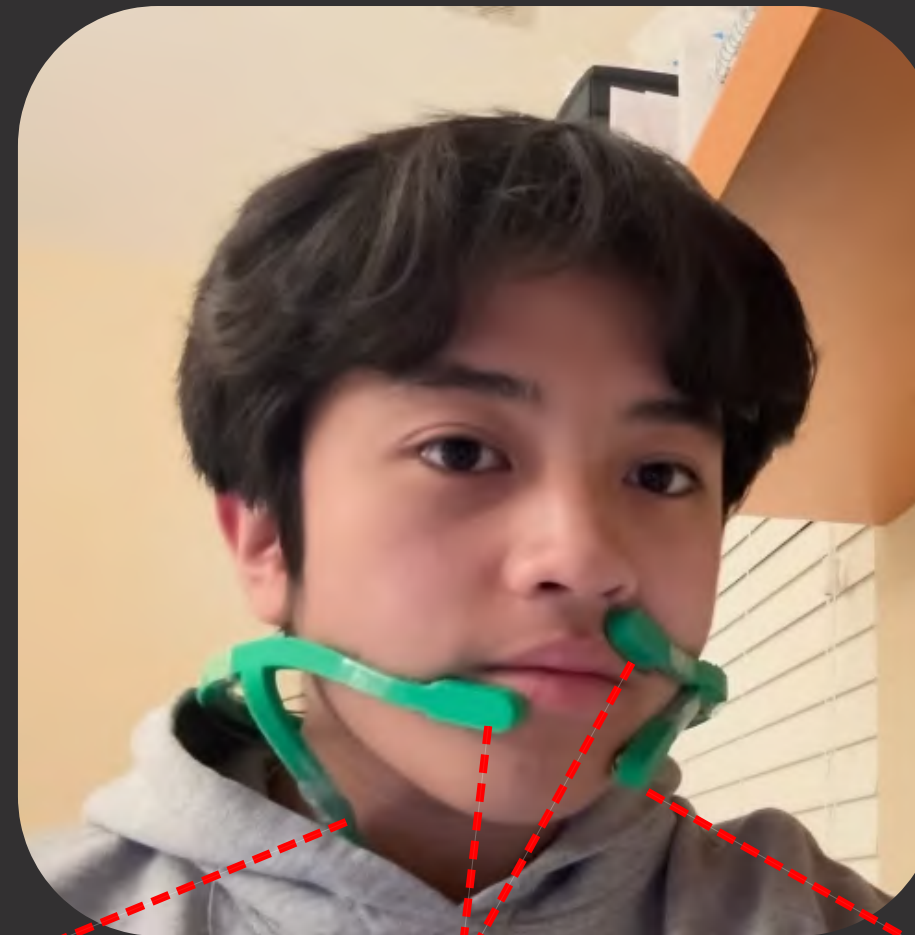
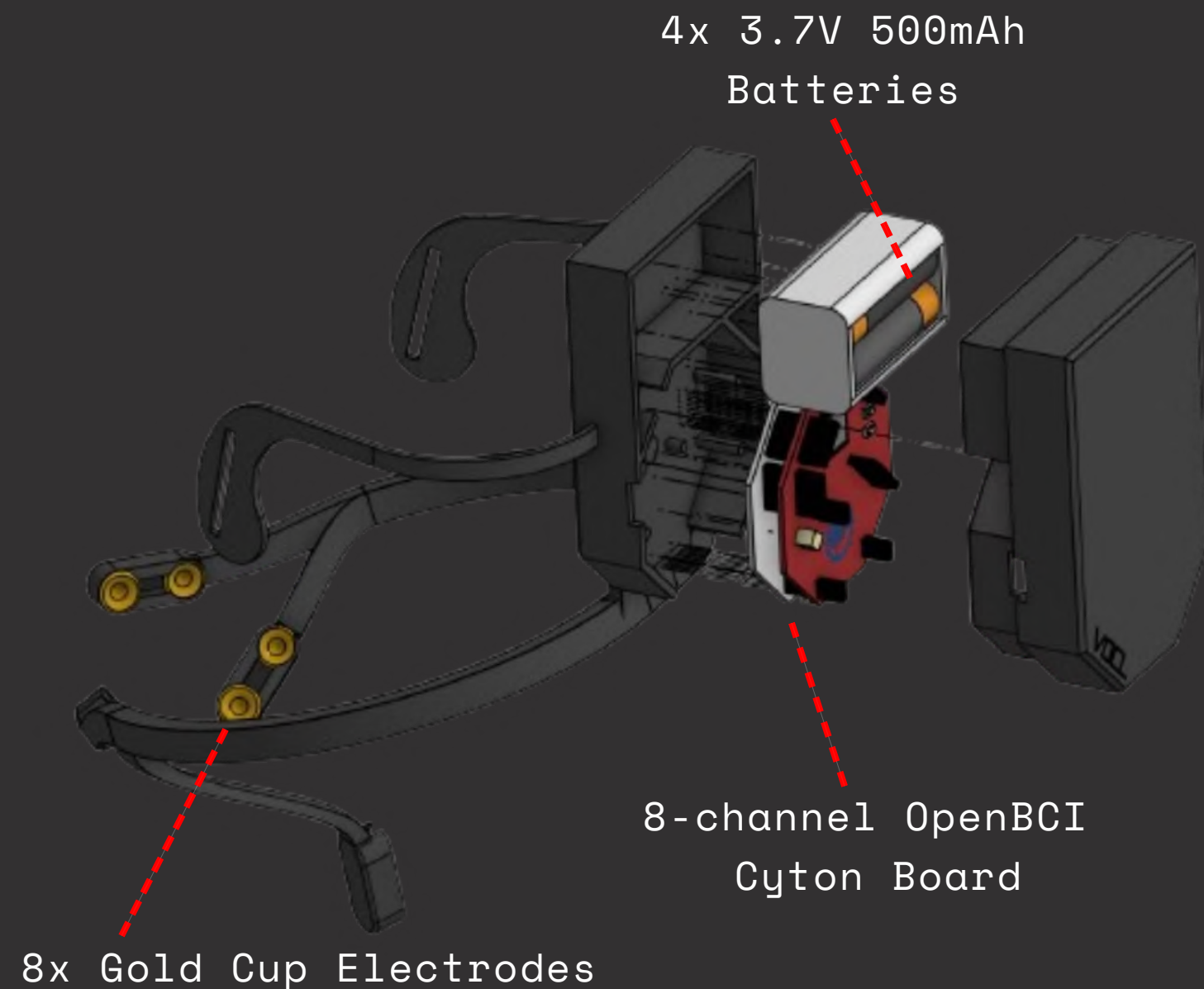
01.
Non-invasive design

02.
Accurate and fast

03.
Affordable for all families

04.
Targets the “Missing Middle”

Hardware



Platysma

Orbicularis Oris
Superior/Inferior

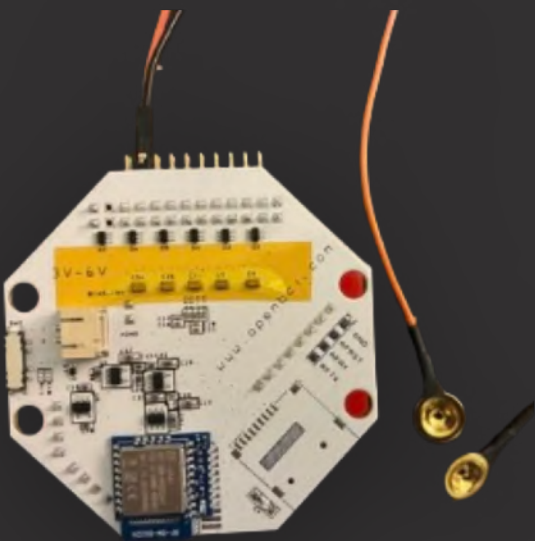
Depressor
Labii

Specifications

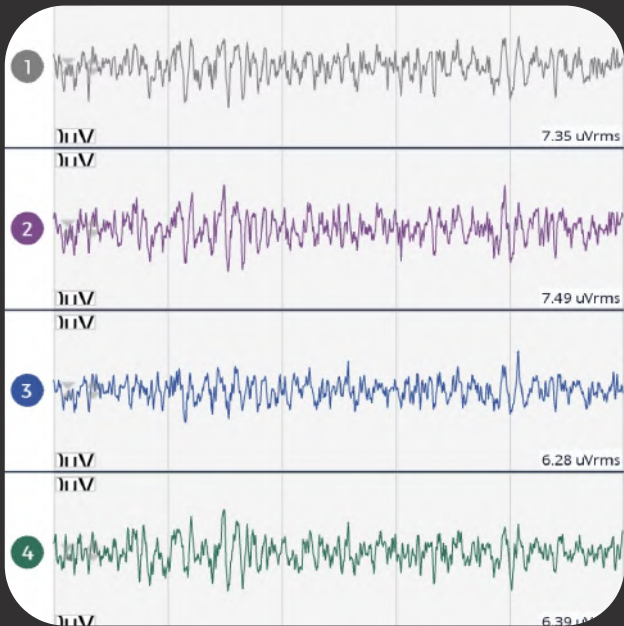
- 3D-Printed **PETG** Frame
- Lightweight at **356.9g**
- **4 Electrode Channels**
- **250Hz** Sampling Rate
- **~8-10 Hours** of Battery Life
- Total BOM: **\$645.50**

System Architecture

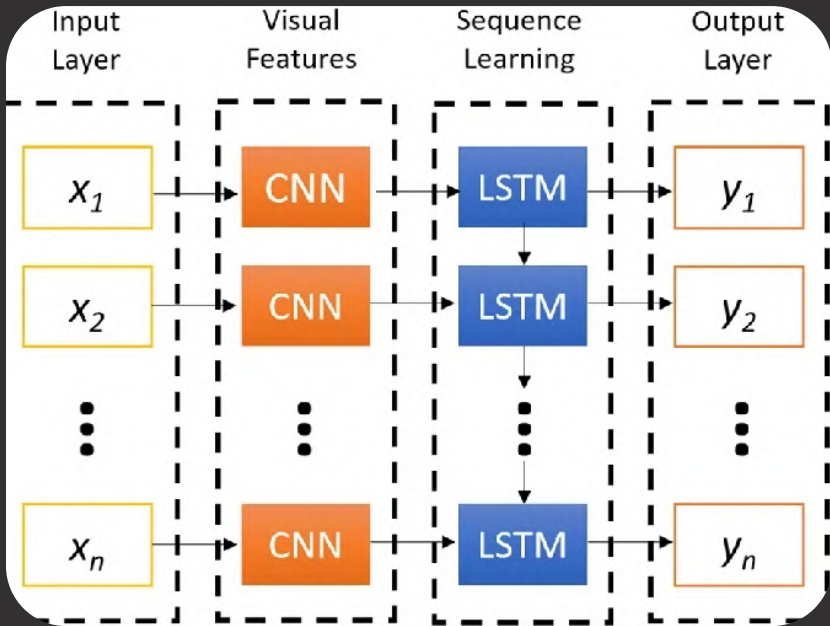
Hardware → Signal Processing → CNN-LSTM Model → LLM Integration



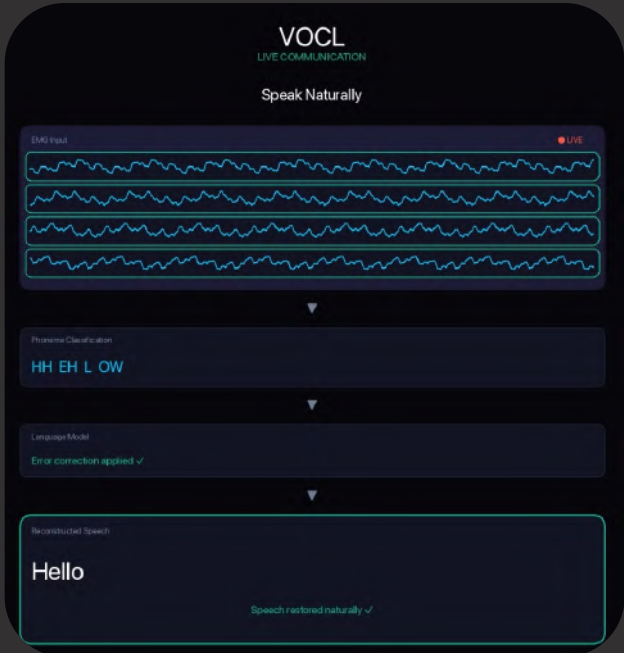
Open BCI Cyton Board,
Gold Cup Electrodes



Facial EMG (fEMG)



CNN-LSTM Diagram



VOCL App Connection
to Headset

Key Results

98.1%

Vocalized Phoneme
Accuracy

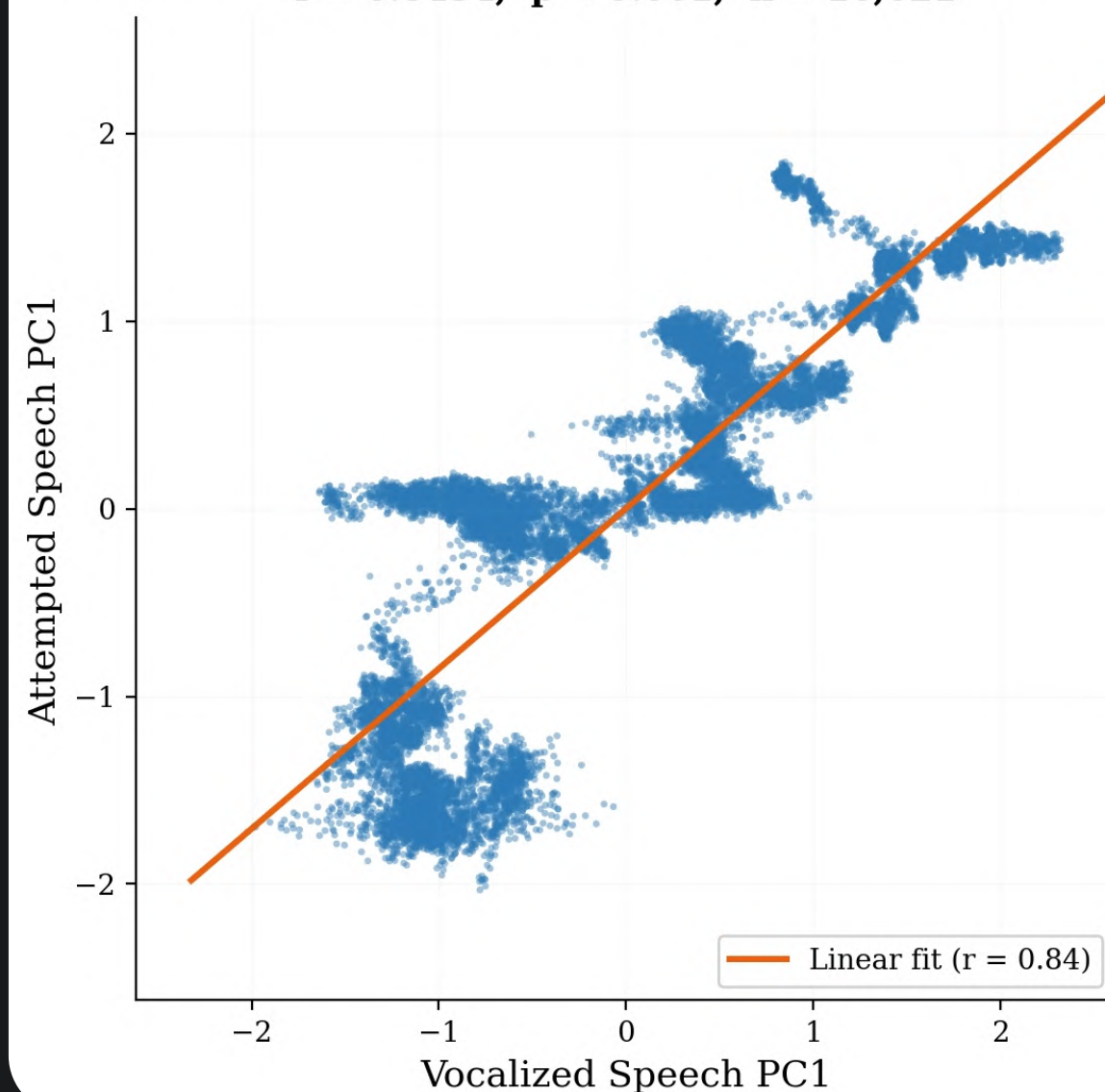
82.0%

Attempted Speech
Accuracy

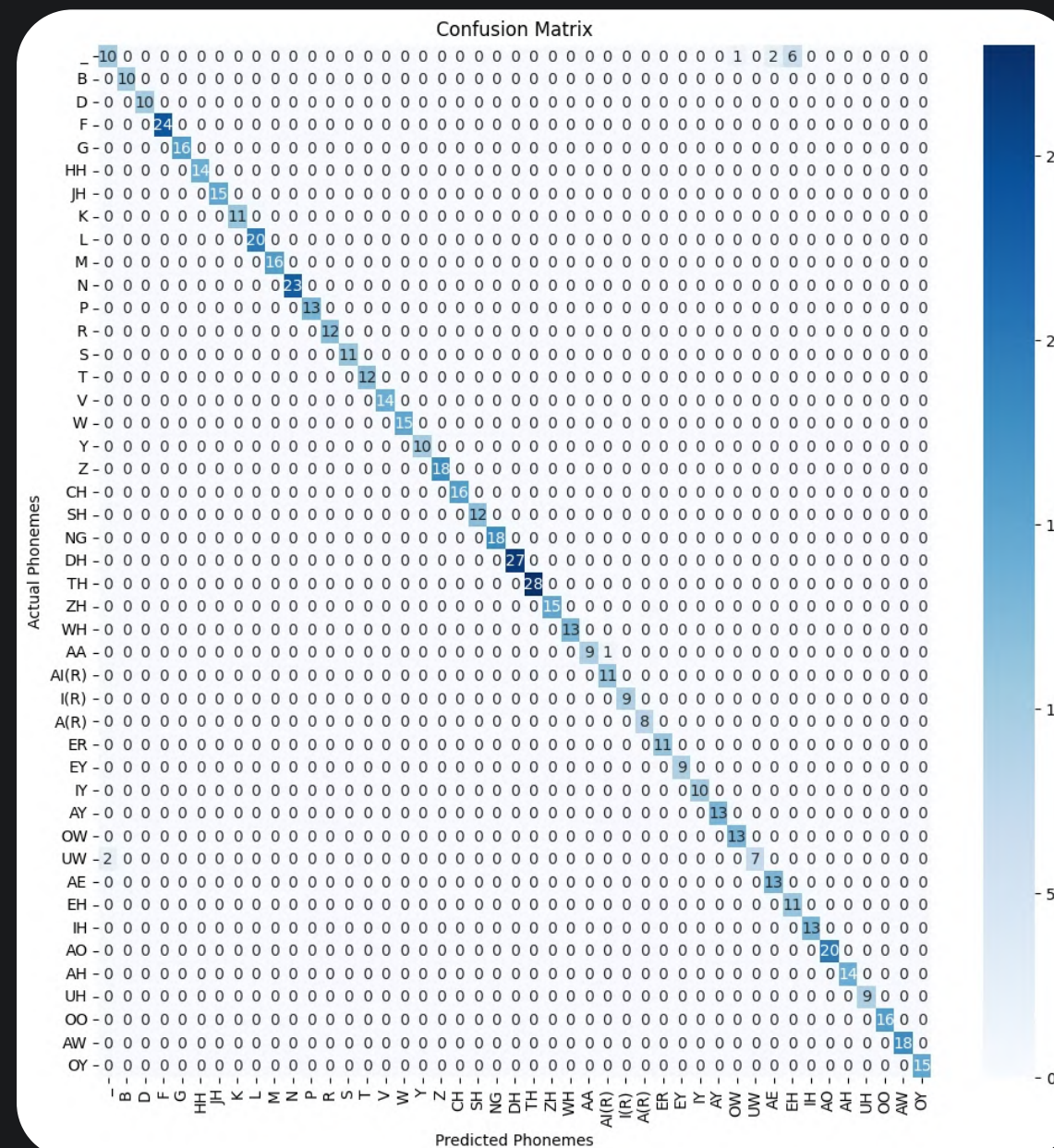
$r=0.84$

Vocalized & Attempted
Speech EMG Correlation

PC1 Correlation Between Vocalized and Attempted Speech
 $r = 0.8434$, $p < 0.001$, $n = 16,621$



Vocalized and Attempted Speech



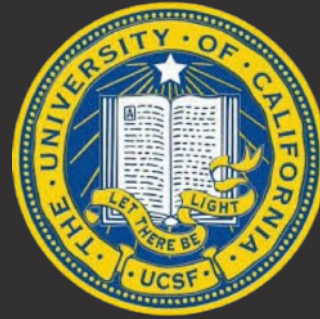
Confusion Matrix (Predicted
vs. Actual Phonemes)

Validation



**Dr. Nicholas
Hatsopoulos**

**2024 Quantrell Award
for Excellence in
Undergraduate
Teaching**



**Dr. David
Moses**

**Top three worldwide in
the International BCI
Award for three
consecutive years**



**Dr. Stephanie
Palmer**

**1997 Rhodes Scholar,
recently named one of
six Schmidt Science
Polymaths**



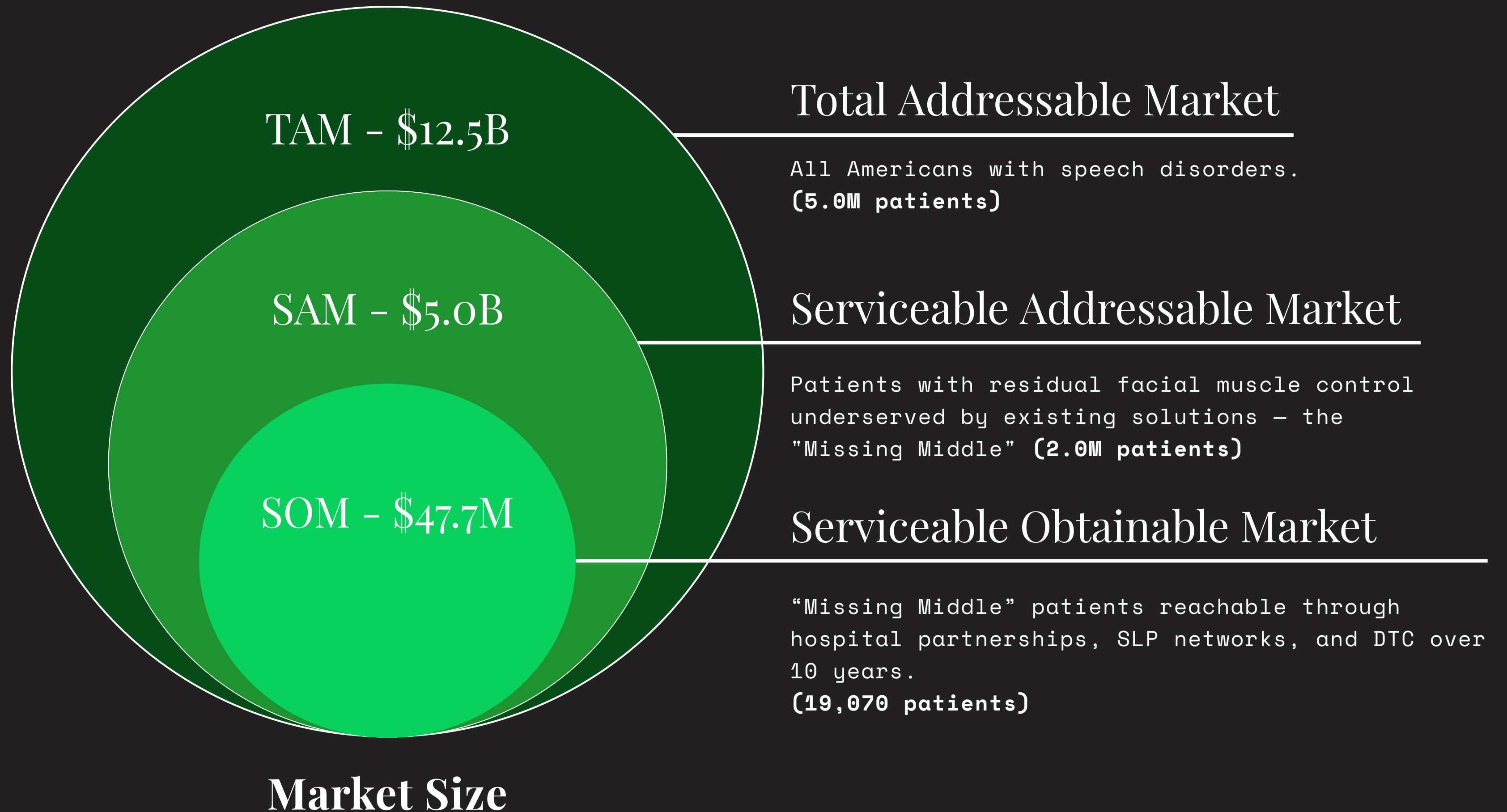
**Dr. Lee
Miller**

**American Institute for
Medical and Biological
Engineering (AIMBE)
College of Fellows**

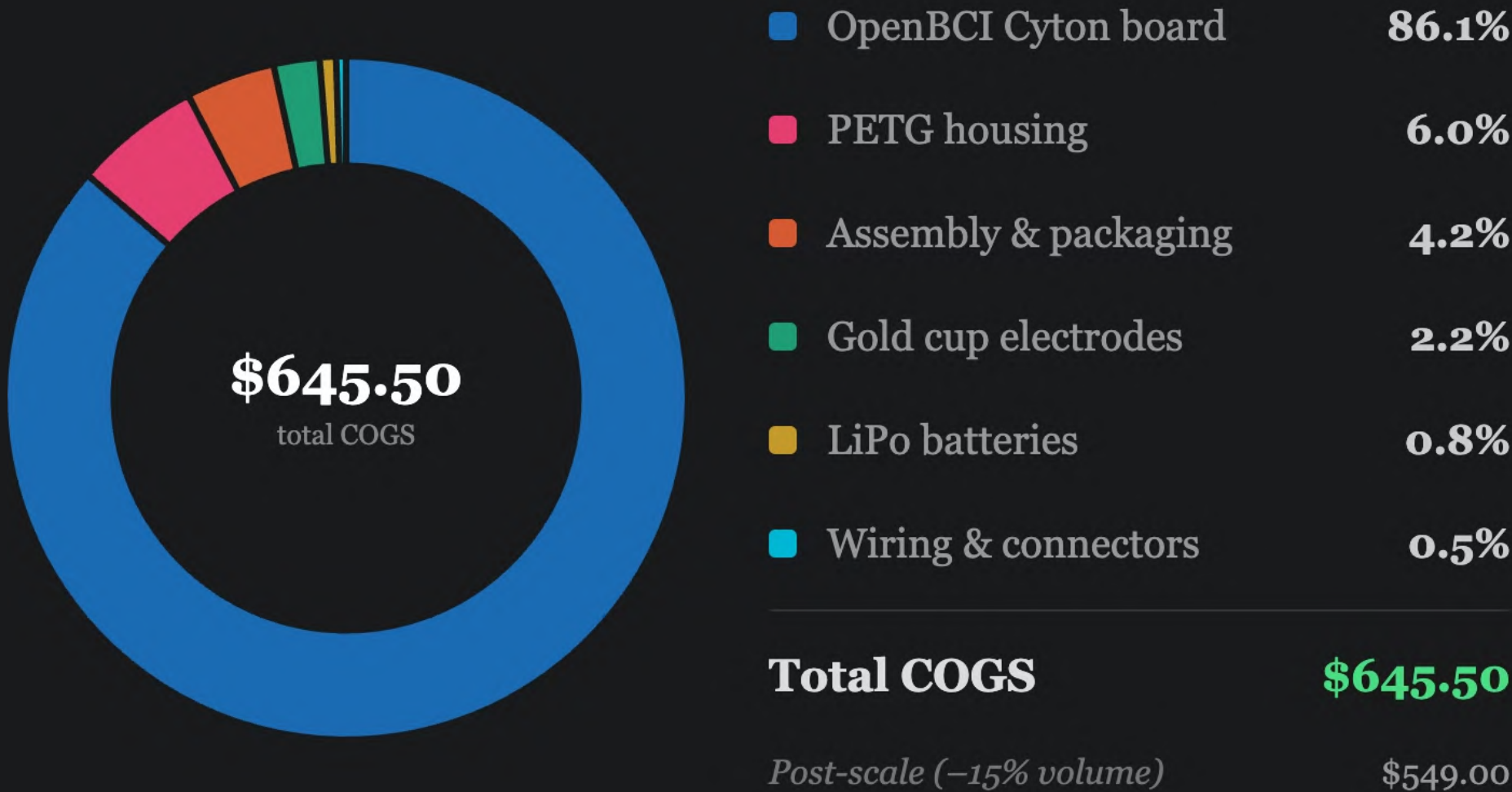




Feature	Traditional AAC	Invasive BCI	VOCL Echo
Price	\$7,000–\$15,000	>\$100,000	\$2,500 (Target) 
Speed	8–15 WPM	15–50+ WPM 	30–50 WPM (Target) 
Non-Invasive?	Yes 	No	Yes 
Hands Free?	No	Yes 	Yes 
Medicare Eligible?	Yes (E2510) 	No (Investigational)	Yes (E2510 Target) 



COGS and BOM



Component	Cost
OpenBCI Cyton 8-channel board	\$555.50
3D-printed PETG housing	\$38.75
Assembly & packaging	\$27.25
Gold cup electrodes (8-pack)	\$14.00
LiPo batteries	\$5.00
Wiring, connectors, misc.	\$3.00
Total Cost of Goods Sold (COGS)	\$645.50
COGS with 15% scale reduction	\$549.00
Gross Margin at \$2,500 ASP	78%

Unit Economics

SELLING PRICE

\$2,500

DTC · B2B ~\$2,100

GROSS MARGIN (POST-SCALE)

78%

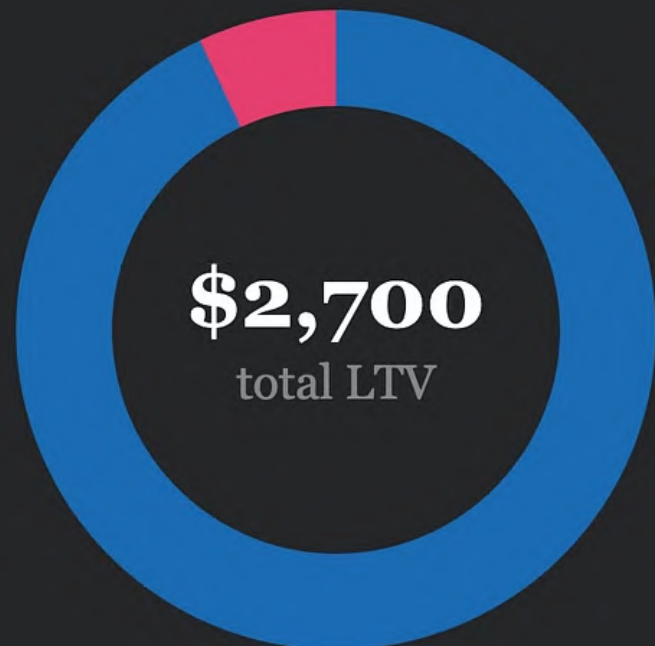
Pre-scale: 74%

LTV : CAC

7.7 : 1

\$2,700 LTV / \$350 CAC

DTC LTV — PER CUSTOMER

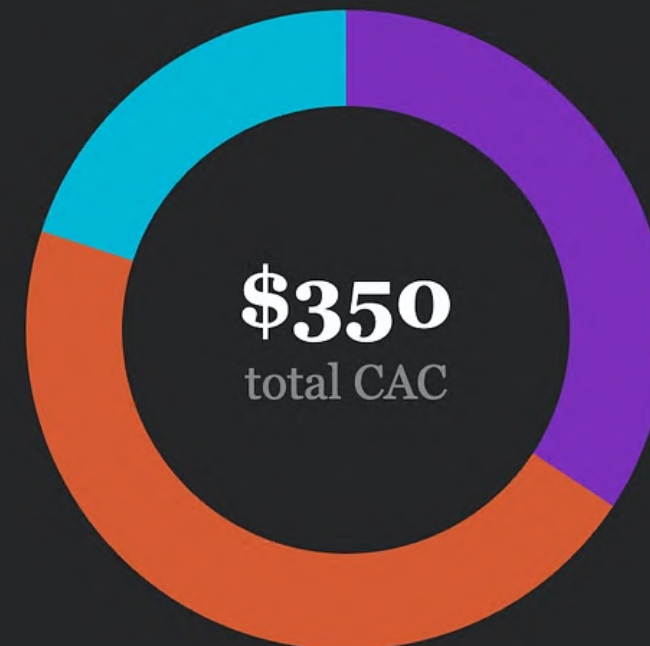


■ Hardware \$2,500

■ Subscriptions \$188

Total LTV **~\$2,700**

CAC — BLENDED DTC / B2B



■ Digital / advocacy \$120

■ Clinical demo + sales \$160

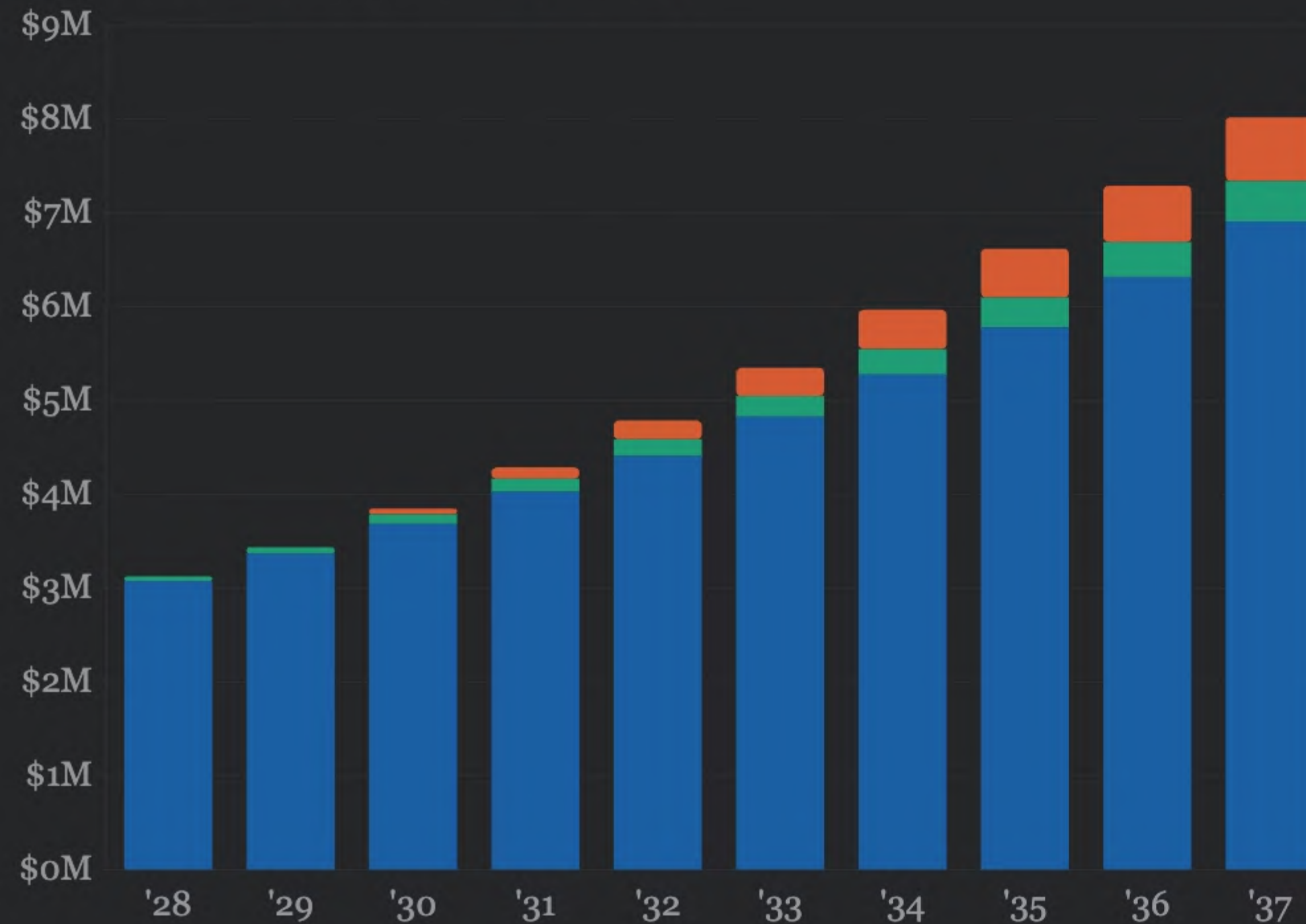
■ Partner development \$70

Total CAC **\$350**

Business Model

REVENUE BY STREAM — BASE CASE (\$M)

■ Hardware ■ Subscriptions ■ Licensing



CAGR

9.4%

Matches U.S. SGD market rate

REGULATORY + REIMBURSEMENT

FDA 510(k)

Medicare E2510 pathway

SCALE TRIGGER

Custom PCB in year 3

COGS -15% → 78% gross margin

Ye-Ye couldn't wait. The next patient shouldn't have to either.



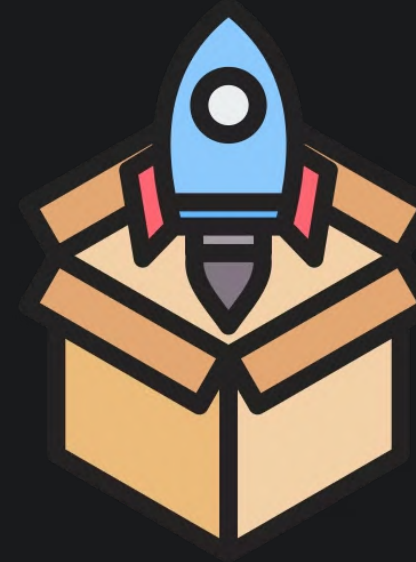
2026

IRB pilot, 15-20
ALS patients, real-
world validation



2027

FDA 510(k)
submission +
manufacturing
partner



2028-2029

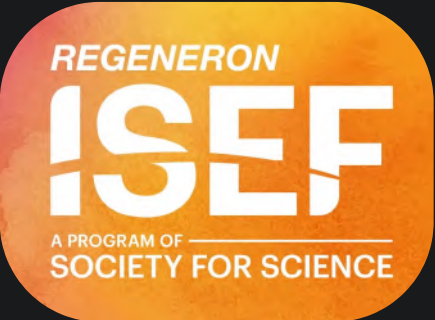
Commercial launch,
DTC + clinical
channel



2030+

Platform expansion
– stroke,
dysarthria,
pediatric

A promise for a better future.





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